Name: ______________________ Date: ____________ Period: ________

Stem-and-Leaf Plot

Stem-and-leaf plots are useful for organizing and analyzing large data sets. In a stem-and-leaf plot, the data are ordered from least to greatest and organized by place value. The digits in the largest place value form the stems. The digits in the least place value form the leaves. A key explains the stems and leaves.

Example 1: Represent the data from the table in a stem-and-leaf plot.

Amount Ian Earned Mowing Lawns per Week ($)			
14	22	30	25
35	44	40	18
25	10	25	15

Step 1: Order data:

10, 14, 15, 18, 22, 25, 25, 25, 30, 35, 40, 44

Step 2: Identify and Label Stems (largest place value)

Tens: 1, 2, 3, 4

Step 3: Identify and Label Leaves (least place value)

Ones

Step 4: Create Key (provides example of the data)

2|5 = $25

Stem	Leaf
1	0 4 5 8
2	2 5 5 5
3	0 5
4	0 4

Why is a key necessary in a stem-and leaf plot?

It clarifies the place values of stem and leaves

Example 2: Interpret data from a stem-and-leaf plot.

The stem-and-leaf plot shows the distance from school that each student in Mr. Romero's class lives from school. Interpret the data.

Distance Students Live From School

Stem	Leaf
0	5 5 9
1	0 0 3 4 5 9
2	0 2 5 7
3	0 2 4 5
4	
5	2

1|3 = 1.3 miles

$$\div 2 \left(\frac{9}{18} = \frac{x}{100} \right) \div 2$$

$x = 50\%$

1. According to the key, what does 1|3 represent? 1.3 miles
2. What is the shortest distance from school? 0.5 miles
3. What is the longest distance from school? 5.2 miles
4. What is the range of the data? 5.2 - 0.5 = 4.7
5. How many data values do I have? 18
6. What is the mode? 0.5 and 1.0
7. What percentage of students live less than 2 miles away from school?
50%

Number of Calories in Selected Snack Foods

Stem	Leaf
24	0 4 4 8
25	0 0 5 7 8
26	4 5
27	5
28	4

24 | 4 = 244 Calories

1. What is the greatest number of calories in the snack food? 284
2. What is the least number of calories in the snack food? 240
3. How many data values do I have?
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